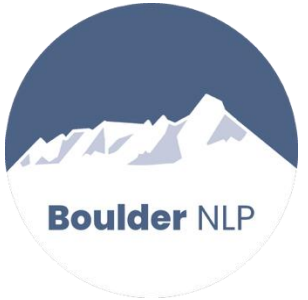
Natural Language Processing

Jim Martin -- CSCI 5832

Lecture 15



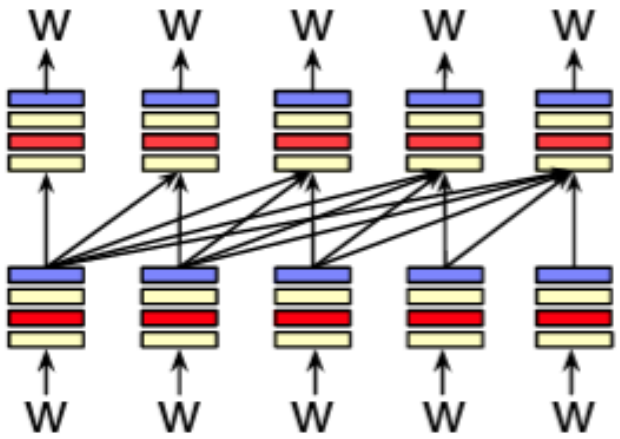
Today

- Quiz 2 is Tuesday 10/27
  - Chapters 5, 6, 7
    - Embeddings
    - Neural Nets
    - Transformers and Pretraining
- Finish Encoder-Decoder Transformers
- Pretraining/Post-Training Recipes

Architectures

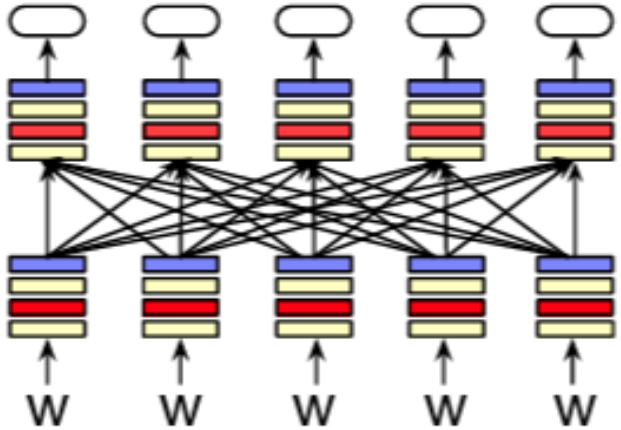
1. Language Models
2. Gradient-based learning
3. Embeddings

GPT family



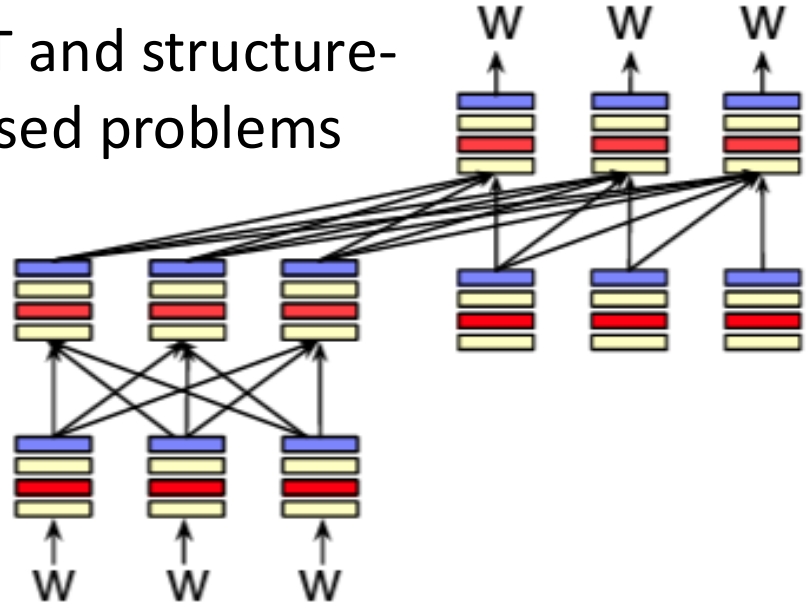
Decoder

BERT family



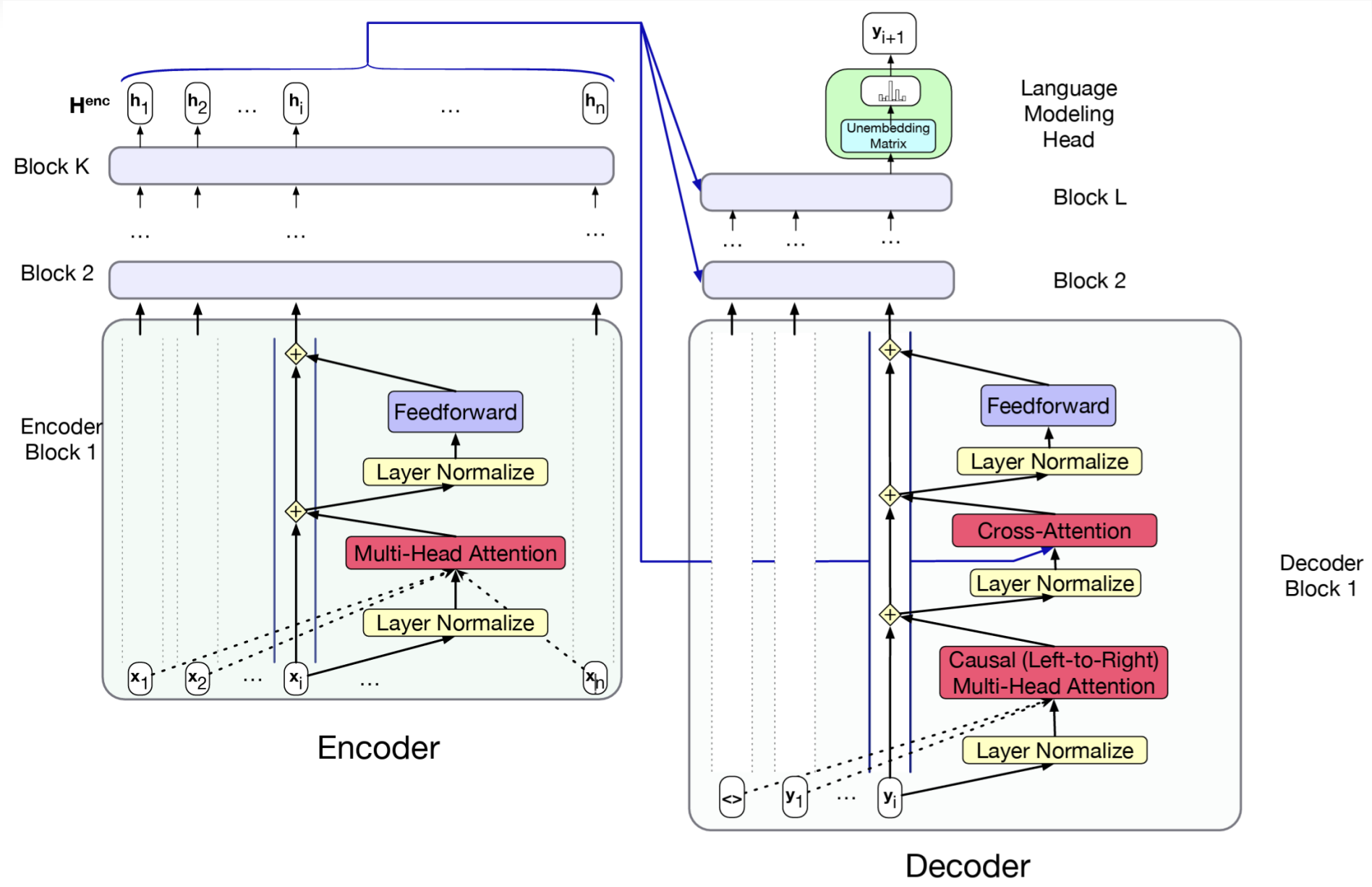
Encoder

MT and structure-
Based problems



Encoder-Decoder

Encoder-Decoder



Cross-Attention

Decoder previous hidden layer

Encoder final hidden layer

$$\mathbf{Q} = \mathbf{H}^{dec[\ell-1]}\mathbf{W}^{\mathbf{Q}};\ \ \mathbf{K} = \mathbf{H}^{enc}\mathbf{W}^{\mathbf{K}};\ \ \mathbf{V} = \mathbf{H}^{enc}\mathbf{W}^{\mathbf{V}}$$

$$\text{CrossAttention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\mathsf{T}}}{\sqrt{d_k}}\right)\mathbf{V}$$

Encoder-Decoder Architecture:Inference

- Input passed to encoder to generate encoder H embeddings.
- Decoder autoregressively generates a token at a time.
- Decoder attention layers look at previous auto-generated outputs
- Decoder cross-attention looks at encoder hidden embeddings.

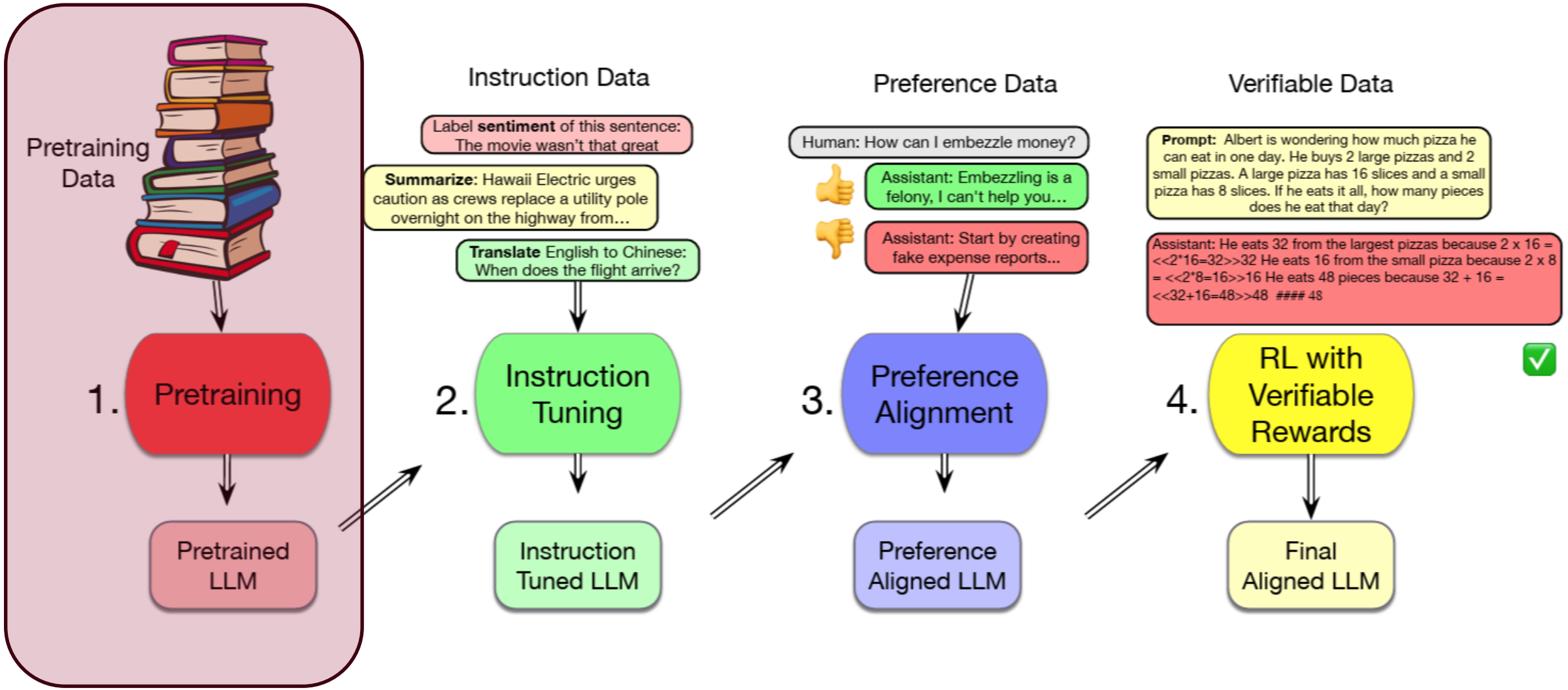
Encoder-Decoder: Training

- Input consists of 2 parts: a source input and a corresponding target output.
- The source is provided to encoder and through that to decoder via cross-attention
- Target is provided to the decoder as in next token prediction (teacher forcing).
- Loss is negative log likelihood (batched) and backpropagated through all the layers (encoder and decoder).

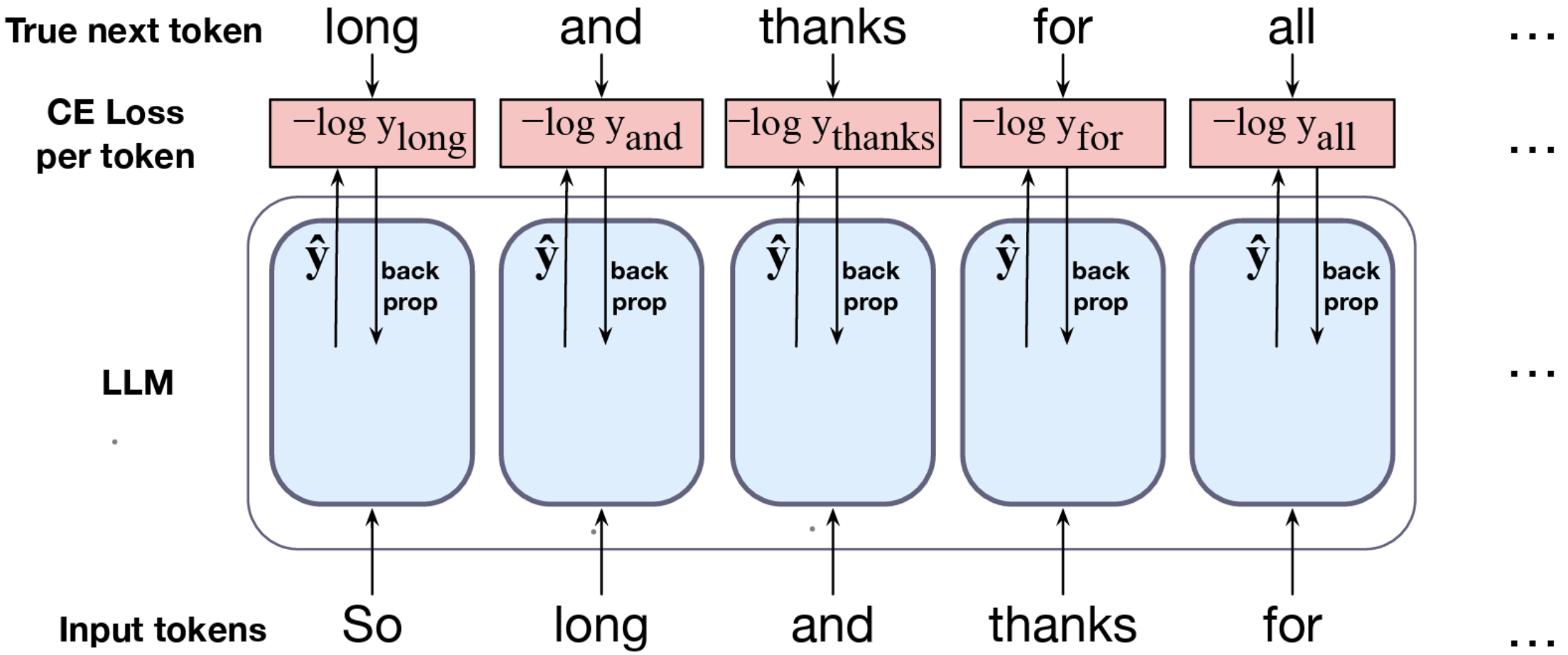
Applications

- Characteristics
  - Length differences between inputs and outputs
  - Different vocabularies between encode input and decoder output.
- Machine translation
  - Source-target bi-texts
- Structure prediction
  - Code generation
    - Problem statement / code output

LLM Pretraining/Post-training Pipeline

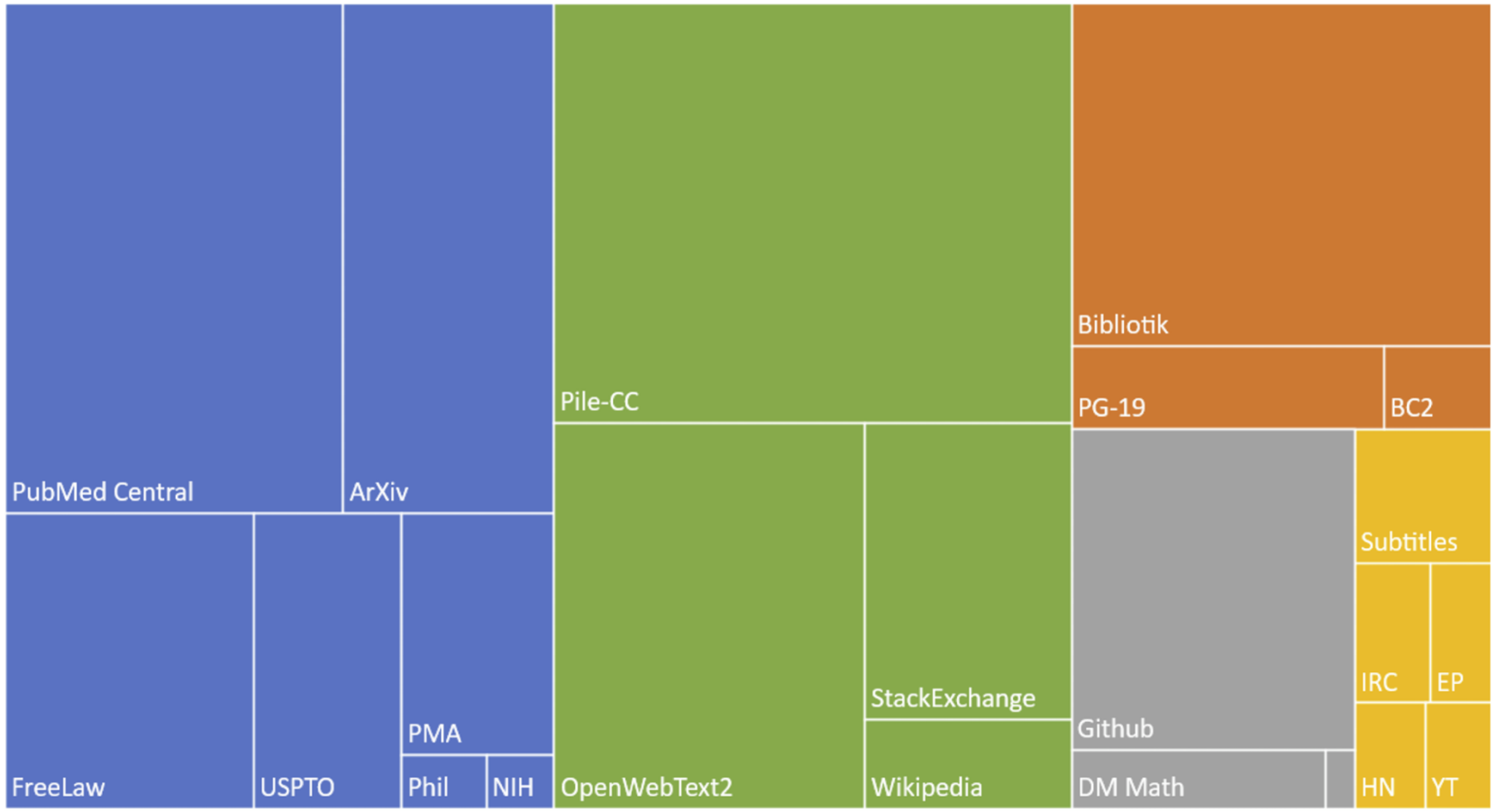


Pre-training



Pretraining: The Pile

- 825 GB of English (mainly)
  - Academic texts
  - Internet scrape
    - Stack exchange
    - Wikipedia
    - Reddit
  - Books corpus
  - Dialog
  - Misc
    - public github



Modern Pretraining Recipes

1. Traditional pretraining as we've been assuming.
2. Mid-training: Same deal, but with higher quality training data.
3. Length training: Same deal but changing the size of the input context
    1. Start with limited context. Increase in size after mid-training. Continue training with high quality data.

Example

Llama 3 (2024) is pretrained with 15T (multilingual) tokens.

	8B	70B	405B
Layers	32	80	126
Model Dimension	4,096	8192	16,384
FFN Dimension	14,336	28,672	53,248
Attention Heads	32	64	128
Key/Value Heads	8	8	8
Peak Learning Rate	3×10^{-4}	1.5×10^{-4}	8×10^{-5}
Activation Function	SwiGLU		
Vocabulary Size	128,000		
Positional Embeddings	RoPE ($\theta = 500,000$)		

Pretraining/Post-training Pipeline

Pretraining Data



1. Pretraining

Pretrained LLM

Instruction Data

Label **sentiment** of this sentence: The movie wasn't that great

Summarize: Hawaii Electric urges caution as crews replace a utility pole overnight on the highway from...

Translate English to Chinese: When does the flight arrive?

2. Instruction Tuning

Instruction Tuned LLM

Preference Data

Human: How can I embezzle money?

👍 Assistant: Embezzling is a felony, I can't help you...

👎 Assistant: Start by creating fake expense reports...

3. Preference Alignment

Preference Aligned LLM

Verifiable Data

Prompt: Albert is wondering how much pizza he can eat in one day. He buys 2 large pizzas and 2 small pizzas. A large pizza has 16 slices and a small pizza has 8 slices. If he eats it all, how many pieces does he eat that day?

Assistant: He eats 32 from the largest pizzas because 2 x 16 = <<2*16=32>>32 He eats 16 from the small pizza because 2 x 8 = <<2*8=16>>16 He eats 48 pieces because 32 + 16 = <<32+16=48>>48 #### 48

✅

4. RL with Verifiable Rewards

Final Aligned LLM

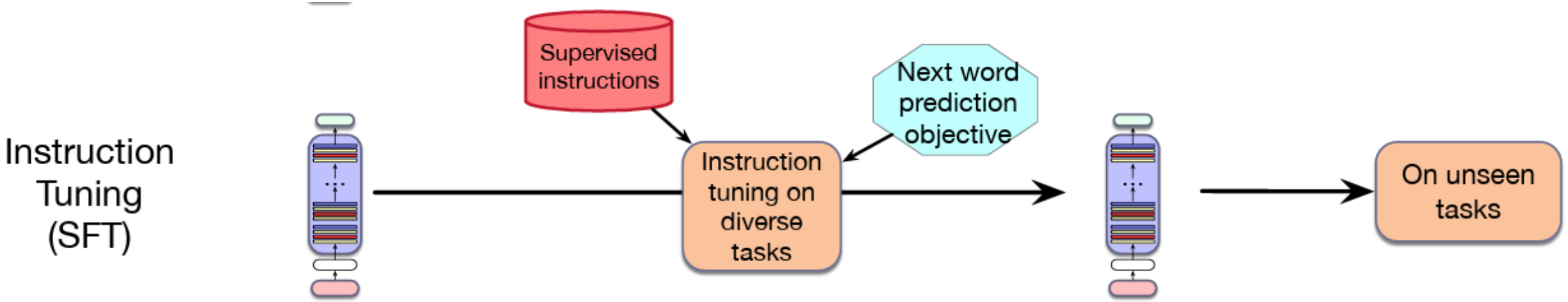
Unfortunately…

By itself, prompting with a large pretrained model doesn't always get what you want (Early GPT-3 examples).

Prompt: Explain the moon landing to a six year old in a few sentences.
Output: Explain the theory of gravity to a 6 year old.

Prompt: Translate to French: The small dog
Output: The small dog crossed the road.

Instruction Tuning



Aka Supervised Finetuning (SFT)

Instruction Data

A prompt-like description of a task to be performed, along with demonstrations of the task. Where can we get that kind of data?

- Have people write the instructions and the data.
- Use available supervised training data from NLP tasks: sentiment, Q/A, translation, etc.
  - These datasets were often crowd-sourced and have detailed guidelines for the workers to follow.
- Use an LLM to generate new answers to existing instructions (replacing crowd workers).
- Use an LLM to generate new tasks with instructions and answers.

Instruction Data: Extractive Q/A

- **Definition:** This task involves creating answers to complex questions, from a given passage. Answering these questions, typically involve understanding multiple sentences. Make sure that your answer has the same type as the "answer type" mentioned in input. The provided "answer type" can be of any of the following types: "span", "date", "number". A "span" answer is a continuous phrase taken directly from the passage or question. You can directly copy-paste the text from the passage or the question for span type answers. If you find multiple spans, please add them all as a comma separated list. Please restrict each span to five words. A "number" type answer can include a digit specifying an actual value. For "date" type answers, use DD MM YYYY format e.g. 11 Jan 1992. If full date is not available in the passage you can write partial date such as 1992 or Jan 1992.
- **Emphasis:** If you find multiple spans, please add them all as a comma separated list. Please restrict each span to five words.
- **Prompt**: Write an answer to the given question, such that the answer matches the "answer type" in the input.
  Passage: { passage}
  Question: { question }

What's the goal?

We don't want to use Instruction Tuning to produce a model that knows how to answer questions, translate, do sentiment, etc. by being tuned on those tasks.

We want a model that learns to follow instructions generically. So that when it encounters new tasks from users it can follow those instructions (meta-learning)

Diverse Instructions

The Super-Natural Instructions Dataset has instructions (and demonstrations) for 1600 tasks (76 overlapping kinds of tasks).

To ensure meta-learning, train and test on different kinds of tasks.



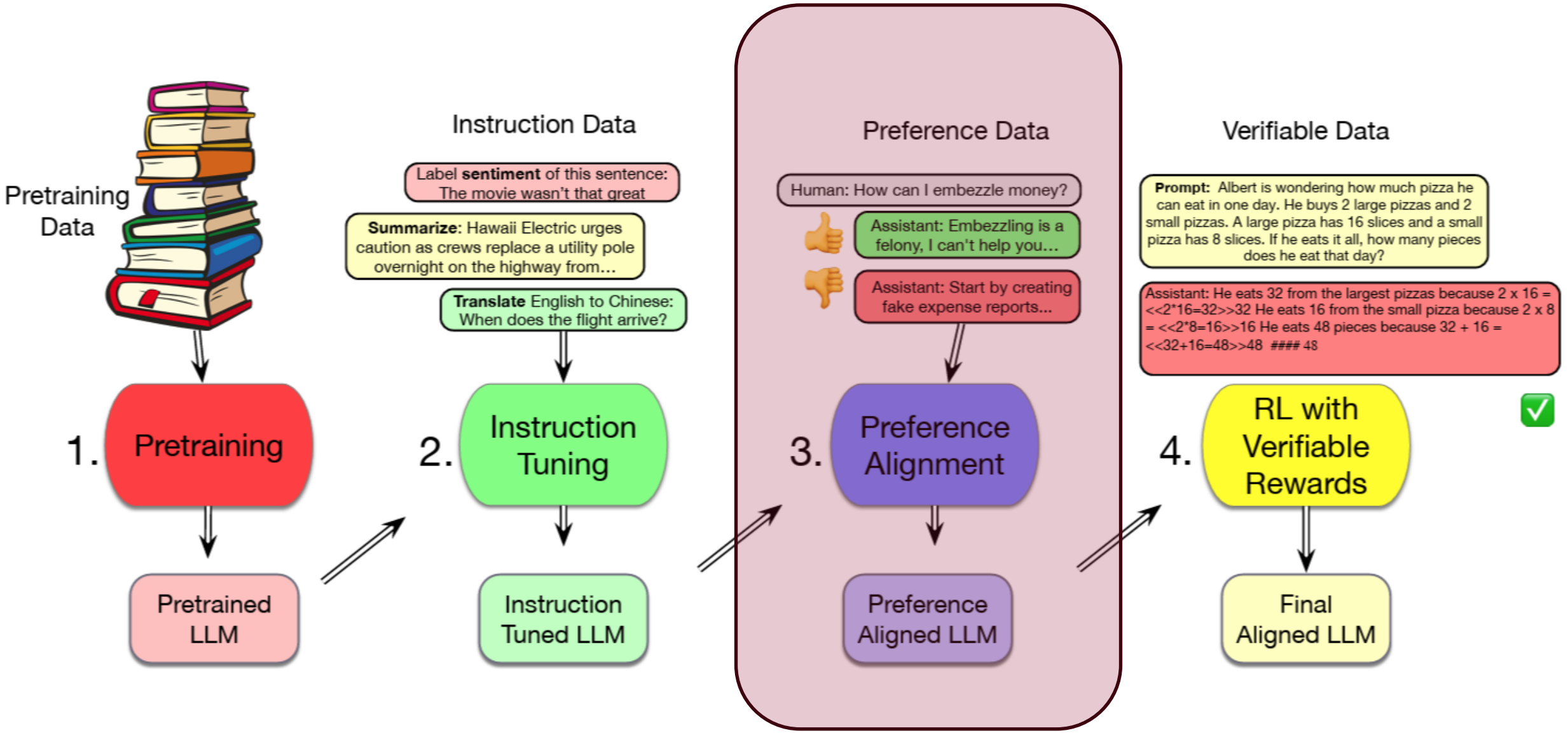
Instruction Tuning Loss

Given a stream of properly tokenized data, the training loss is just a variation on the standard language modeling loss.

$$\mathcal{L}_{\mathrm{SFT}}(\theta) = -\sum_{(x,y)\in\mathcal{I}} \sum_{t=1}^{T} \log p_\theta(y_t | x, y_{<t})$$

Where only *y* contributes to the loss, and y corresponds to the tokens in the assistant response. The prompt x has no impact since we're learning to respond, not to come up with prompts.

Pretraining/Post-training Pipeline



Instruction Data: Extractive Q/A

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  Question: { question }

Vs. Preferences

Prompt: I've heard garlic is a great natural antibiotic. Does it help with colds?
Chosen: It can be helpful against colds, but may make you stink.
Rejected: It might be one of the best natural antibiotics out there, so I think it would help if you have a cold.

Prompt: What is malaria?
Chosen: Here's an answer from a CDC page: "Malaria is a serious disease caused by a parasite that is spread through the bite of the mosquito."
Rejected: I don't know what malaria is.